# Few-Shot Continual Learning for 3D Brain MRI with Frozen Foundation Models

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***Abstract*— Foundation models pretrained on large 3D medical imaging corpora must often be adapted to a sequence of downstream tasks from few labeled examples, without revisiting earlier tasks' data. We study this few-shot continual setting on 3D brain MRI with a frozen 3D U-Net adapted by one Low-Rank Adaptation (LoRA) module and output head per task, over a sequence of glioma segmentation (BraTS 2023), stroke-lesion segmentation (ISLES 2022) and brain-age regression (IXI) at 32 labeled volumes each, under a subject-grouped held-out protocol with whole-volume inference and three seeds. Three results stand out. First, where a task needs global context and encoder capacity—brain age from whole-head input—LoRA is the only frozen method that learns it at all, reaching MAE $10.22 \pm 0.82$ yr ($R^2 = 0.40$) against a constant-predictor baseline of 14.4 yr, beating full fine-tuning of 14M parameters by 1.98 yr ($p = 3\times10^{-9}$) with 56K trainable, while frozen-encoder linear probing and BitFit emit a constant. Second, where a large task-specific decoder already exists—segmentation—linear probing is marginally better (Dice $0.789 \pm 0.005$ vs. $0.777 \pm 0.004$) at 90× the trainable parameters, and the two are equal by 64 shots. Third, a naive reading of "frozen backbone" that also freezes the 66-parameter output projection costs 0.22 Dice. We further report two negative results: EWC, LwF and experience replay all forget *more* than unregularised fine-tuning in this regime, and the pretrained checkpoint we use is not better than random initialisation for the frozen backbone ($-0.009$ Dice, $p = 3\times10^{-9}$), so the mechanism we demonstrate is parameter-efficient adaptation rather than transfer from pretraining. Code, splits, per-case predictions and the corrected evaluation protocol are released.**

***Index Terms*— Few-shot learning Continual learning LoRA 3D brain MRI Parameter-efficient fine-tuning Brain age estimation.**

## I. Introduction

MEDICAL imaging AI increasingly relies on foundation models pretrained on large-scale datasets [1], [2]. When deployed in clinical settings, these models must adapt to multiple tasks—e.g., tumor segmentation and brain age estimation—often with limited labeled data and without access to prior task samples due to privacy or storage constraints. In practice, radiology departments may introduce new analysis tasks over time (e.g., adding brain age prediction to an existing tumor workflow) with only a small batch of newly annotated cases. This scenario motivates *few-shot continual learning*: learning new tasks sequentially from few examples while retaining performance on previously learned tasks.

Sequential fine-tuning of shared backbone parameters typically causes *catastrophic forgetting* [3], [4]: when the model is updated for a new task, performance on prior tasks degrades. Regularization- and distillation-based remedies such as EWC [3] and LwF [5] soften this at the cost of tuning effort and only approximate protection. An alternative is to *freeze* the pretrained backbone and adapt only small task-specific modules. Parameter-efficient fine-tuning methods such as LoRA [6], [7] inject low-rank matrices into selected layers, training a small fraction of parameters while preserving the original weights.

We study a frozen FOMO-style pretrained 3D U-Net [1] adapted with task-specific LoRA modules to a sequence of heterogeneous tasks—glioma segmentation on BraTS 2023, stroke-lesion segmentation on ISLES 2022, and brain-age regression on IXI—from 32 labeled volumes each. The premise is simple: because the backbone and prior adapters are never modified, earlier tasks cannot be forgotten. The interesting questions are what this costs in accuracy relative to training more of the network, and whether a frozen backbone is adequate for every task type. The answer turns out to depend on *where* the task needs capacity, and to run in opposite directions for segmentation and for regression.

Answering those questions required first fixing how they are measured. In the course of this work we found that the evaluation pipeline commonly used for this setting—and used in an earlier version of this manuscript—reports on the split used for model selection, scores a single random $64^3$ crop per case, pools voxels across cases into one Dice, and lets longitudinal timepoints of one patient span train and test. We replaced it with subject-grouped train/validation/test splits, whole-volume sliding-window inference, and per-case Dice and 95th-percentile Hausdorff distance (HD95). Every number below is from that protocol.

Our findings and contributions are:

- **Where capacity placement matters, LoRA wins outright.** Given whole-head input, brain age is learnable at 32 shots: LoRA with a trainable head reaches MAE $10.22 \pm 0.82$ yr and $R^2 = 0.40$, beating the constant predictor by 4.21 yr ($p = 5\times10^{-31}$) and full fine-tuning by 1.98 yr ($p = 3\times10^{-9}$). Linear probing of a 513-

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parameter head on the frozen encoder, and BitFit, both collapse to a constant (prediction SD 0.09 yr against a true 16.5). The same task from $64^3$ patches is unlearnable for every method—the failure was the input, not the task.

- **Where a large task head already exists, LoRA trades a little accuracy for 90× fewer parameters.** On glioma segmentation, LoRA + head at rank 16 reaches Dice $0.777 \pm 0.004$ against $0.789 \pm 0.005$ for full-decoder linear probing (paired $-0.012$, $p = 3 \times 10^{-10}$) while training 93.8K rather than 8.45M parameters; at 64 shots the two are equal (0.811 both).
- **A 66-parameter configuration detail dominates.** Leaving the segmentation output projection frozen, as "frozen backbone" naively implies, costs $0.217$ Dice ($p < 10^{-130}$) and triples seed variance.
- **Forgetting-free is shared, and the standard remedies do not deliver it.** Adapter isolation and frozen-encoder probing both have BWT exactly zero over a three-task sequence in which full fine-tuning loses $0.55$ Dice; EWC, LwF and replay all forget *more* than unregularised fine-tuning ($-0.39$, $-0.56$, $-0.37$ vs. $-0.33$).
- **A negative result on pretraining.** The pretrained checkpoint is not better than random initialisation for the frozen backbone ($-0.009$ Dice, $p = 3 \times 10^{-9}$). We therefore claim parameter-efficient adaptation, not transfer.
- **A corrected evaluation protocol and code**, with tests that guard the label-handling and split-leakage failures we encountered.

## II. Related Work

### A. Foundation Models for Medical Imaging

Large-scale pretrained models have transformed medical image analysis. The FOMO initiative [1] provides 3D foundation models pretrained on diverse medical imaging datasets via masked autoencoding objectives, yielding general-purpose representations for downstream adaptation. UNETR [8] introduced Vision Transformers into 3D medical segmentation by replacing CNN encoders with self-attention, demonstrating strong performance on multi-organ and brain tumor segmentation. More recently, SAM-Med3D [9] extended the Segment Anything paradigm to volumetric data, and domain-specific foundation models have emerged for pathology [10] and radiology. These models share a common deployment challenge: adapting to multiple downstream tasks efficiently while preserving the pretrained representations, particularly when labeled data is scarce.

### B. Continual Learning in Medical Imaging

Continual learning addresses the problem of sequentially learning new tasks without forgetting previously acquired knowledge [4]. Classical approaches fall into three categories: (i) *regularization-based* methods such as EWC [3] and Synaptic Intelligence [11], which penalize changes to important parameters; (ii) *replay-based* methods such as iCaRL [12], which store or generate exemplars from prior tasks; and (iii) *distillation-based* methods such as LwF [5], which use the previous model's outputs as soft targets. In medical imaging, continual learning has been explored primarily for class-incremental segmentation: LifeLonger [13] benchmarks continual disease classification across dermatology datasets, while Lifelong nnU-Net [14] standardizes continual organ segmentation. However, most prior work considers *class-incremental* settings with homogeneous outputs (e.g., adding new organ classes). Our work addresses *task-incremental* continual learning across *heterogeneous* output types—segmentation and regression—which presents distinct challenges since the task heads and loss functions differ fundamentally.

### C. Parameter-Efficient Fine-Tuning

Parameter-efficient fine-tuning (PEFT) methods adapt large pretrained models by training only a small subset of parameters. LoRA [6] injects trainable low-rank decompositions into weight matrices, achieving competitive performance with full fine-tuning at a fraction of the parameter cost. Alternative PEFT approaches include Adapters [15], which insert bottleneck modules between layers; Prefix-Tuning [16], which prepends learnable tokens to the input; and BitFit [17], which tunes only bias terms. In the context of continual learning, PEFT methods offer a natural mechanism for task isolation: PackNet [18] iteratively prunes and freezes network capacity per task, while Piggyback [19] learns binary masks over a shared backbone. Our approach is most closely related to adapter-isolation strategies but applies LoRA's low-rank structure to 3D convolutions, which is directly compatible with pretrained U-Net architectures. A recurring detail in the PEFT literature, and the one that turns out to matter most here, is that the task-specific output layer is always trained alongside the low-rank updates; a pretrained projection cannot be expected to map onto a new label space.

## III. Method

### A. Problem Formulation

We consider continual learning with sequential task arrival. Let $f_\theta$ denote a pretrained 3D U-Net backbone (frozen) and $\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_K$ a sequence of tasks. For each task $k$, we have access to only $N_k$ labeled few-shot samples ($N_k = 32$ throughout) drawn from task-specific distribution $\mathcal{D}_k = \{(x_i^{(k)}, y_i^{(k)})\}_{i=1}^{N_k}$. We assume *no replay*: when training on $\mathcal{T}_k$, we cannot access data from $\mathcal{T}_{<k}$. The optimization objective for task $k$ is:

$$\min_{\phi_k, h_k} \frac{1}{N_k} \sum_{i=1}^{N_k} \mathcal{L}_k\big(h_k(f_\theta(x_i^{(k)}; \phi_k)),\ y_i^{(k)}\big) \tag{1}$$

where $\phi_k$ denotes the LoRA adapter parameters for task $k$, $h_k$ the task-specific head, and $\mathcal{L}_k$ the task loss. Crucially, $\theta$ remains frozen throughout; only $\phi_k$ and $h_k$ are optimized. The goal is to learn $\{\phi_k, h_k\}_{k=1}^K$ such that performance on all seen tasks remains high after training on $\mathcal{T}_K$.

We evaluate two tasks: (1) tumor segmentation (3D binary mask) and (2) brain age estimation (regression). Each task has modality-specific inputs (BraTS: T2w, T2-FLAIR, T1c; IXI: T1, T2) and a distinct output head $h_k$. Because the modality counts differ, the first convolution of the encoder cannot be

shared and is task-specific; all remaining encoder weights (39 of 40 tensors) are shared and frozen.

Figure 1 illustrates the overall pipeline: a frozen pretrained backbone, task-specific LoRA adapters, and task heads. At inference for task $k$, we use backbone $f_\theta$ + adapter $\phi_k$ + head $h_k$; prior adapters and the backbone remain frozen. Task identity is assumed known at inference.

### B. LoRA for 3D Convolutions

Low-Rank Adaptation (LoRA) [6] injects trainable low-rank matrices into a frozen weight matrix $W \in \mathbb{R}^{d\times k}$:

$$W' = W + \Delta W, \quad \Delta W = \frac{\alpha}{r} B \cdot A, \quad A \in \mathbb{R}^{r\times k},\ B \in \mathbb{R}^{d\times r} \tag{2}$$

with rank $r \ll \min(d, k)$ and scaling factor $\alpha$. Only $A$ and $B$ are trained; $W$ stays frozen. The scaling ratio $\alpha/r$ controls the magnitude of the low-rank perturbation relative to the pretrained weights.

For 3D convolutions in a U-Net, we apply LoRA via parallel $1 \times 1 \times 1$ convolutions acting on the channel dimension:

$$\text{Conv3d}'(x) = \text{Conv3d}(x; W) + \frac{\alpha}{r}\,\text{Conv1}(x; B) \circ \text{Conv1}(x; A) \tag{3}$$

where Conv1 denotes $1 \times 1 \times 1$ convolution, $A \in \mathbb{R}^{r\times C_{\text{in}}}$ projects to rank-$r$ space, and $B \in \mathbb{R}^{C_{\text{out}}\times r}$ projects back. This formulation is memory-efficient and compatible with standard Conv3d layers [20].

We use $\alpha = 16$, evaluate $r \in \{4, 8, 16\}$ (Sec. V-C) and report $r = 16$, and target every Conv3d in both encoder and decoder for segmentation; encoder-only suffices for regression. For a single Conv3d layer with input channels $C_{\text{in}}$ and output channels $C_{\text{out}}$, the number of LoRA parameters is:

$$|\phi_{\text{layer}}| = r \cdot (C_{\text{in}} + C_{\text{out}}) \tag{4}$$

which is substantially smaller than the original $C_{\text{out}} \times C_{\text{in}} \times D \times H \times W$ parameters. Summed over all targeted layers at $r = 16$, the adapter for segmentation has 93,810 parameters against a 22.63M-parameter U-Net (0.41%), and the adapter for regression has 55,841 parameters against the 14.16M-parameter encoder-plus-head (0.39%); at $r = 8$ the counts are 46,872 and 28,177.

### C. Continual Learning via Adapter Isolation

Each task $k$ receives a *dedicated LoRA adapter* $\phi_k$. The full training and inference procedure is summarized in Algorithm 1. The trainable parameter set for task $k$ is strictly disjoint from all other tasks:

$$\Phi_k = \{\phi_k, h_k\}, \quad \Phi_i \cap \Phi_j = \emptyset \ \ \forall\, i \neq j \tag{5}$$

Because $\theta$ and $\{\Phi_{<k}\}$ are frozen when optimizing $\Phi_k$, the performance of any prior task $j < k$ is invariant to training on $\mathcal{T}_k$. Formally, let $R_{t,j}$ denote the performance of task $j$ after training up to task $t$. Then:

$$R_{t,j} = R_{j,j} \quad \forall\, t \geq j \implies \text{BWT}_j = R_{K,j} - R_{j,j} = 0 \tag{6}$$

This zero-forgetting property holds *by construction*, without regularization or replay. It is not, however, unique to adapters: any scheme that leaves shared weights untouched—including linear probing on a frozen encoder—satisfies Eq. (6) equally. What distinguishes the schemes is how many parameters they must train to reach a given accuracy, which is what Section V measures. The total model storage after $K$ tasks is:

$$\text{Storage} = |\theta| + \sum_{k=1}^{K} (|\phi_k| + |h_k|) \tag{7}$$

which grows linearly in $K$ with a small per-task overhead since $|\phi_k| \ll |\theta|$.

**Algorithm 1** Continual Learning via LoRA Adapter Isolation

**Require:** Pretrained backbone $f_\theta$; task sequence $\mathcal{T}_1, \ldots, \mathcal{T}_K$; subject-grouped train/val/test splits per task; LoRA rank $r$; epochs $E$
**Ensure:** Trained adapters $\{\phi_k\}_{k=1}^K$ and heads $\{h_k\}_{k=1}^K$

```
Freeze all parameters of f_θ
for k = 1 to K do
    Initialize LoRA adapter φ_k with rank r (Kaiming A, zero B)
    Initialize task head h_k (output projection for segmentation; pooled regressor for regression); mark trainable
    if k > 1 then
        Freeze all previous adapters φ_1, ..., φ_{k-1} and heads h_1, ..., h_{k-1}
    end if
    for epoch = 1 to E do
        for each mini-batch (x, y) ∈ D_k^train do
            ŷ ← h_k(f_θ(x; φ_k))
            Update φ_k, h_k on L_k(ŷ, y) with AdamW
        end for
        Score D_k^val (deterministic centre crop); keep the best epoch
    end for
    Score the kept checkpoint once on D_k^test with whole-volume inference
end for
```

### D. Task Heads and Losses

**Task 1 (Segmentation).** The full U-Net decoder produces a voxel-wise two-class prediction. LoRA is applied to every Conv3d in encoder and decoder. The head $h_1$ is the decoder's final $1\times1\times1$ projection from 32 channels to the 2 output classes (66 parameters), which is trained from its pretrained initialization. We stress this because a natural but wrong reading of "frozen backbone" is to freeze this layer too; Section V-B quantifies the consequence. The training loss is cross-entropy plus soft Dice on the foreground probability:

$$\mathcal{L}_{\text{seg}} = \text{CE}(p, g) + 1 - \frac{2\sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon} \tag{8}$$

where $p_i$ is the softmax foreground probability at voxel $i$, $g_i$ the binary label, and $\epsilon = 10^{-8}$.

**Task 2 (Regression).** For brain age estimation we use the encoder only: its deepest feature map is global-average-pooled

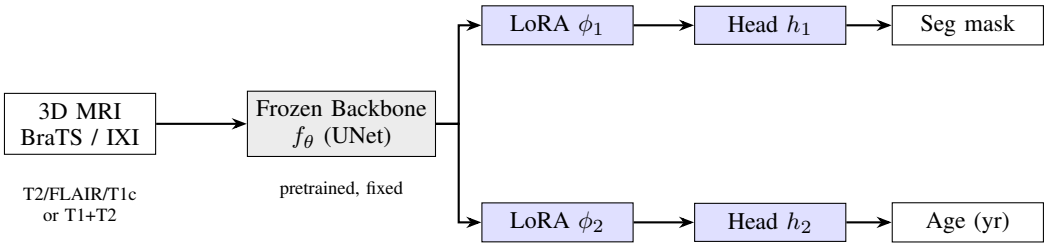



Fig. 1. Framework: frozen pretrained backbone $f_\theta$ with task-specific LoRA adapters $\phi_k$ and heads $h_k$. Gray = frozen; blue = trainable per task. At inference for T1 (resp. T2): backbone + $\phi_1$ + $h_1$ (resp. $\phi_2$ + $h_2$). For segmentation, $h_1$ is the $1\times1\times1$ output projection of the U-Net decoder (66 parameters); leaving it frozen is the single largest source of error we observed (Sec. V-B).

and passed through a two-layer fully connected head with a single scalar output. LoRA targets the encoder's Conv3d layers. The loss is mean squared error in years.

**Optimization.** All methods use AdamW (learning rate $10^{-3}$, weight decay $10^{-5}$) with cosine annealing to $10^{-6}$ over 100 epochs, batch size 2, gradient-norm clipping at 1.0, and LoRA dropout 0.1. Training patches are random $64^3$ crops; no other augmentation is used. Hyperparameters are identical across methods and were not tuned per method.

## IV. Experiments

### A. Datasets

**Notation.** T1 denotes tumor segmentation, T2 brain-age regression, and T3 stroke-lesion segmentation. The main comparison uses the sequence T1→T2; Section V-E also reports the sequence T1→T3→T2.

**T1: BraTS 2023 Glioma** [21]. We use the 1251 labeled volumes of the training release, with channels T2w, T2-FLAIR and T1c and a binary whole-tumor mask derived from the expert annotation. The challenge's validation release ships without segmentations and is excluded; an earlier pipeline had written those 219 volumes with all-zero masks, which are indistinguishable from a tumor-free label and had contaminated 17% of the test set. Several subjects contribute more than one longitudinal timepoint; these are kept in the same split.

**T2: IXI** [22]. Of the 577 subjects with both T1 and T2 scans, 559 have a usable age in the IXI demographics table (15 have no table entry, 3 have a missing age) and are used; none is imputed. Test-set ages span 20.0–86.3 yr (mean 50.0, SD 16.5).

**T3: ISLES 2022.** Ischemic-stroke lesion segmentation with channels DWI, ADC and FLAIR (FLAIR resampled onto the DWI grid) and the expert lesion mask; 247 of the 250 cases are used, three being excluded because their mask emptied under 1 mm resampling. Lesions are small (median ≈775 voxels), which makes this a harder segmentation task than T1 and one drawn from a single cohort, so unlike an earlier infarct-*classification* design that contrasted ISLES with IXI controls, it carries no acquisition confound.

**T2 (whole-head variant).** Because patch input proved to be the limiting factor for regression (Sec. V-D), we also prepare each IXI subject as a single $96^3$ cube: the volume is cropped to its non-zero content and resampled, so one forward pass sees the entire head. The task, labels, splits and metrics are otherwise identical, and models are run with a $96^3$ window, which makes the crop the identity and evaluation a single window.

**Data use.** BraTS 2023 was obtained from Synapse (syn51156910) under its Data Use Agreement; IXI is publicly available at `https://brain-development.org/ixi-dataset/`; ISLES 2022 is distributed under CC BY 4.0 (Zenodo 7153326). No additional ethics approval was required.

**Preprocessing.** Volumes are resampled to 1 mm isotropic and z-scored per channel over non-zero voxels.

### B. Splits and Evaluation Protocol

For each task and seed we draw a subject-grouped three-way split: a training set of exactly $N_k = 32$ volumes, a validation set of 60 subjects used only for epoch selection, and a held-out test set of up to 250 subjects (275 volumes for T1, 250 for T2, all 155 remaining subjects for T3) on which every number in this paper is reported. No subject appears in more than one split. Splits are persisted so that all methods are scored on identical cases, which is what makes the paired comparisons below valid.

**Segmentation** is scored on the whole volume by Gaussian-weighted sliding-window inference ($64^3$ windows, 25% overlap), followed by argmax. We report per-case Dice and per-case HD95 in voxel units, averaged over cases. **Regression** is scored by averaging the prediction over a fixed $2\times2\times2$ grid of $64^3$ windows. We report MAE in years, $R^2$, and the standard deviation of the predictions, the last because it exposes a model that has collapsed to a constant regardless of MAE.

**Backward transfer.** For the sequential baselines, after each later task we re-score every earlier task's test set with that task's own head on the now-updated encoder and report $\mathrm{BWT}_j = R_{K,j} - R_{j,j}$ in Dice; in the three-task sequence we also report the intermediate value after T3. For adapter-isolated methods this quantity is zero by construction and is not re-measured.

### C. Backbone

`unet_b` from FOMO [1], initialized from `fomo25_mmunetvae_pretrained.ckpt`. All 40 encoder tensors except the modality-specific input convolution load from the checkpoint; the decoder loads 59 of 82 tensors, the remainder being normalization parameters. The same checkpoint is loaded at the start of every task for every method.

TABLE I
TRAINABLE PARAMETERS PER METHOD. THE COMPARISON IN TABLE II SPANS FOUR ORDERS OF MAGNITUDE OF TRAINABLE BUDGET.

| | T1 segmentation | | T2 regression | |
|---|---|---|---|---|
| Method | trainable | share | trainable | share |
| LoRA (enc+dec), $r$=8 | 46,872 | 0.207% | 28,177 | 0.199% |
| LoRA + head, $r$=8 | 46,938 | 0.207% | 28,177 | 0.199% |
| LoRA + head, $r$=16 | 93,810 | 0.414% | 55,841 | 0.393% |
| Sequential Linear | 8,446,306 | 37.4% | 513 | 0.004% |
| Sequential FT | 22,582,978 | 100% | 14,136,321 | 100% |

### D. Compared Methods

All methods share the backbone, data, splits, optimizer and schedule; they differ only in which parameters are trainable (Table I).

- **LoRA (enc+dec).** Rank-8 LoRA on every Conv3d; output projection frozen. This is the configuration of the earlier version of this work.
- **LoRA + head (proposed).** As above, plus the 66-parameter output projection. Reported at rank 16; ranks 4 and 8 appear in the ablation of Sec. V-C.
- **Sequential Linear.** Encoder frozen; every downstream parameter trainable. For segmentation this is the entire decoder (8.45M parameters); for regression, the 513-parameter head. One model carried through the sequence.
- **Sequential FT.** All parameters trainable; one model carried through the sequence. The forgetting reference.
- **BitFit** [17]. PEFT comparator below LoRA's budget: bias terms plus the output projection (6.4K parameters), task-isolated like LoRA.
- **EWC** [3], **LwF** [5], **experience replay**. Full fine-tuning plus a forgetting-mitigation term: a diagonal-Fisher penalty ($\lambda = 10^3$), encoder-feature distillation from the previous task's model (weight 0.5), and the same distillation on a 100-sample buffer of previous-task volumes, respectively.
- **Random-initialisation control.** LoRA + head ($r$=16) with the backbone initialised randomly instead of from the checkpoint, to test whether pretraining contributes anything (Sec. V-F).

Per-task oracles from an earlier version were run under the pre-audit protocol only and are omitted rather than reported on a different footing.

### E. Implementation and Resources

Experiments run on a single NVIDIA RTX 3090. Measured peak GPU memory is 1.67 GB during segmentation training (batch 2) and 4.16 GB during whole-volume evaluation (batch 8 windows); regression peaks at 0.69 GB and 2.53 GB respectively. One segmentation run trains in 46 min and evaluates 275 volumes in about 25 min; regression trains in 14 min. Code, splits and per-case predictions are released with the paper.

## V. RESULTS

All results are on held-out test splits at $N_k = 32$, reported as mean ± SD over three seeds (42–44); each seed draws its own subject-grouped split, and all methods share the split within a seed. Paired comparisons pool per-case differences across seeds ($n = 820$ for BraTS, 750 for brain age). All 66 runs completed; none failed.

### A. Main Comparison

Table II gives the comparison on the two-task sequence and Table III the paired tests.

### B. The Output Projection Accounts for Most of the LoRA Gap

At rank 8, with the decoder's final $1\times1\times1$ projection frozen, LoRA on every other layer reaches Dice $0.538 \pm 0.082$ and HD95 $135 \pm 14$; the spread across seeds (0.452–0.648) reflects how much the outcome depends on whether the pretrained projection happens to be usable for the new label. Training those 66 additional parameters lifts it to $0.754 \pm 0.029$ and HD95 $48.6 \pm 5.0$—a paired gain of $+0.217$ Dice ($p = 1\times10^{-134}$) at a budget that changes from 46,872 to 46,938, and a threefold reduction in seed variance. Mechanically the pretrained projection maps decoder channels onto the pretraining objective's output space, and rank-8 perturbations upstream cannot re-map it onto a new label space. Every PEFT recipe trains the new head; a "frozen backbone" should not be read as including it.

### C. Accuracy per Trainable Parameter, and Rank

Sequential Linear remains more accurate than LoRA + head at rank 16, but narrowly: Dice $0.789\pm0.005$ vs. $0.777\pm0.004$, a paired $-0.012$ ($p = 3\times10^{-10}$), and HD95 41.9 vs. 43.2, with equal seed stability. It reaches that accuracy by training the full 8.45M-parameter decoder; LoRA + head trains 93.8K, a factor of 90 fewer, for a 1.5% relative Dice cost. Full fine-tuning is better still, by $+0.021$ over LoRA + head, at 240× the budget and at the cost of forgetting (Sec. V-E).

Table IV varies the rank. Rank 4 is not worse than rank 8 ($+0.007$, $p = 9\times10^{-6}$) at half the parameters, so rank 8 sits in a flat region; rank 16 gains $+0.024$ over rank 8 ($p = 5\times10^{-40}$) and cuts the seed SD from 0.029 to 0.004. BitFit, at 6.4K parameters, is far too small for this task (0.441) and marks the lower end of the budget curve.

The gap also closes with data (Table V): at 16 shots linear probing leads by 0.092, at 32 by 0.012, and at 64 the two are equal (0.811 both). The parameter advantage of LoRA is therefore paid for only in the smallest-data regime.

### D. Brain Age: Input Scope Decides, and LoRA Wins

From $64^3$ patches, no method learns brain age: all ten configurations in Table II land within 14.4–14.7 yr of the constant predictor (14.4 yr) with $R^2 \approx 0$, including full fine-tuning of every parameter. A $64^3$ window covers a few percent of a $240\times240\times180$ head, and the cues for age—ventricular volume, cortical thinning, global atrophy—are not local.

Re-running the task with the whole head resampled to a single $96^3$ cube changes the picture entirely (Table VI). LoRA + head reaches MAE $10.22 \pm 0.82$ yr with $R^2 = 0.40$,

TABLE II

HELD-OUT TEST RESULTS, $N_k = 32$, MEAN ± SD OVER SEEDS 42–44, SEQUENCE T1→T2. T1 = BRATS GLIOMA SEGMENTATION (275 VOLUMES, 250 SUBJECTS PER SEED); T2 = IXI BRAIN AGE FROM $64^3$ PATCHES (250 SUBJECTS). "T1 AFTER T2" IS T1 RE-SCORED AFTER THE MODEL HAS LEARNED T2; BWT IS ITS CHANGE. CONSTANT-PREDICTOR MAE FOR T2 IS ≈14.4 YR.

| | T1 segmentation | | T2 brain age (patch) | | Forgetting | | |
|---|---|---|---|---|---|---|---|
| Method | Dice↑ | HD95↓ | MAE (yr)↓ | $R^2$ | T1 after T2 | BWT | Trainable |
| *Frozen backbone (task-isolated; BWT = 0 by construction)* | | | | | | | |
| BitFit | $0.441 \pm 0.034$ | $72.7 \pm 0.5$ | $14.60 \pm 0.07$ | $-0.05$ | = T1 | 0.000 | 6,434 |
| LoRA (enc+dec), $r$=8 | $0.538 \pm 0.082$ | $135.1 \pm 14.0$ | $14.51 \pm 0.39$ | $-0.06$ | = T1 | 0.000 | 46,872 |
| LoRA + head, $r$=4 | $0.761 \pm 0.023$ | $50.1 \pm 3.3$ | $14.35 \pm 0.38$ | $-0.03$ | = T1 | 0.000 | 23,502 |
| LoRA + head, $r$=8 | $0.754 \pm 0.029$ | $48.6 \pm 5.0$ | $14.40 \pm 0.31$ | $-0.01$ | = T1 | 0.000 | 46,938 |
| LoRA + head, $r$=16 | $0.777 \pm 0.004$ | $43.2 \pm 0.7$ | $14.72 \pm 0.37$ | $-0.09$ | = T1 | 0.000 | 93,810 |
| Sequential Linear | $0.789 \pm 0.005$ | $41.9 \pm 1.3$ | $14.64 \pm 0.12$ | $-0.06$ | $0.789 \pm 0.005$ | 0.000 | 8,446,306 |
| *Backbone updated per task* | | | | | | | |
| Sequential FT | $0.798 \pm 0.025$ | $40.2 \pm 4.9$ | $14.52 \pm 0.10$ | $-0.03$ | $0.470 \pm 0.084$ | $-0.328 \pm 0.062$ | 22,582,978 |
| EWC | $0.791 \pm 0.029$ | $41.9 \pm 5.6$ | $14.59 \pm 0.10$ | $-0.05$ | $0.403 \pm 0.146$ | $-0.388 \pm 0.173$ | 22,582,978 |
| LwF | $\mathbf{0.804 \pm 0.022}$ | $\mathbf{38.5 \pm 4.1}$ | $14.64 \pm 0.20$ | $-0.05$ | $0.240 \pm 0.088$ | $-0.564 \pm 0.084$ | 22,582,978 |
| Replay | $0.772 \pm 0.041$ | $46.3 \pm 8.6$ | $14.54 \pm 0.17$ | $-0.04$ | $0.404 \pm 0.143$ | $-0.368 \pm 0.108$ | 22,582,978 |

TABLE III

PAIRED DIFFERENCES IN PER-CASE T1 DICE, POOLED OVER THREE SEEDS ($n = 820$; BOOTSTRAP 95% CI; WILCOXON SIGNED-RANK). L+H$_r$ = LORA + HEAD AT RANK $r$; SL = SEQUENTIAL LINEAR; FT = SEQUENTIAL FT; L$_8$ = LORA (ENC+DEC) WITH THE OUTPUT PROJECTION FROZEN; RI = RANDOM-INITIALISED BACKBONE.

| Contrast | ΔDice | 95% CI | $p$ |
|---|---|---|---|
| L+h$_{16}$ − SL | −0.012 | [−0.015, −0.008] | $3\times10^{-10}$ |
| FT − L+h$_{16}$ | +0.021 | [+0.017, +0.025] | $1\times10^{-33}$ |
| L+h$_{16}$ − L+h$_8$ | +0.024 | [+0.020, +0.027] | $5\times10^{-40}$ |
| L+h$_8$ − L$_8$ | +0.217 | [+0.207, +0.228] | $1\times10^{-134}$ |
| L+h$_{16}$ − L+h$_{16}$(RI) | −0.009 | [−0.012, −0.007] | $3\times10^{-9}$ |

TABLE IV

LORA RANK ABLATION (LORA + HEAD) AND THE BITFIT COMPARATOR, MEAN ± SD OVER SEEDS 42–44. SEQUENTIAL LINEAR SHOWN FOR REFERENCE.

| Config | Dice↑ | HD95↓ | MAE (yr) | Trainable (T1) |
|---|---|---|---|---|
| BitFit | $0.441 \pm 0.034$ | $72.7 \pm 0.5$ | 14.60 | 6,434 |
| $r = 4$ | $0.761 \pm 0.023$ | $50.1 \pm 3.3$ | 14.35 | 23,502 |
| $r = 8$ | $0.754 \pm 0.029$ | $48.6 \pm 5.0$ | 14.40 | 46,938 |
| $r = 16$ | $\mathbf{0.777 \pm 0.004}$ | $\mathbf{43.2 \pm 0.7}$ | 14.72 | 93,810 |
| Seq. Linear | $0.789 \pm 0.005$ | $41.9 \pm 1.3$ | 14.64 | 8,446,306 |

beating the constant predictor by 4.21 yr per subject (paired, $p = 5\times10^{-31}$) and full fine-tuning by 1.98 yr ($p = 3\times10^{-9}$). Sequential Linear and BitFit do *not* recover: their predictions still have a standard deviation of 0.09 yr against a true 16.5, i.e. they remain constant.

This is the paper's clearest result and it runs opposite to segmentation. Brain age needs the *encoder* representation to change—a 513-parameter head on frozen features cannot express it, and bias-only adaptation cannot either—while 56K well-placed low-rank parameters can, and do so better than releasing all 14M. Where segmentation already has a large task-specific decoder to train, regression has nothing comparable, and that is exactly where adapter capacity becomes necessary rather than merely economical.

TABLE V

SHOT ABLATION, BRATS DICE, MEAN ± SD OVER SEEDS 42–44.

| Method | $N_k = 16$ | $N_k = 32$ | $N_k = 64$ |
|---|---|---|---|
| LoRA + head, $r$=16 | $0.655 \pm 0.015$ | $0.777 \pm 0.004$ | $0.811 \pm 0.005$ |
| Sequential Linear | $0.747 \pm 0.031$ | $0.789 \pm 0.005$ | $0.811 \pm 0.019$ |

TABLE VI

BRAIN AGE WITH WHOLE-HEAD INPUT ($\mathbf{96^3}$) VERSUS $\mathbf{64^3}$ PATCHES, MEAN ± SD OVER SEEDS 42–44, 250 TEST SUBJECTS PER SEED. PRED. SD IS THE STANDARD DEVIATION OF THE PREDICTED AGES; THE TRUE SD IS 16.5 YR, SO A VALUE NEAR ZERO DENOTES A CONSTANT PREDICTOR.

| | whole head ($96^3$) | | | patch |
|---|---|---|---|---|
| Method | MAE (yr)↓ | $R^2$ ↑ | pred. SD | MAE |
| LoRA + head, $r$=16 | $\mathbf{10.22\pm0.82}$ | $\mathbf{+0.40\pm0.09}$ | 11.21 | 14.72 |
| Sequential FT | 12.20±0.17 | +0.15±0.04 | 10.60 | 14.52 |
| Sequential Linear | 14.65±0.09 | −0.06±0.03 | 0.09 | 14.64 |
| BitFit | 14.63±0.13 | −0.06±0.02 | 0.09 | 14.60 |
| constant predictor | 14.39 | 0.00 | 0 | 14.39 |

## E. Forgetting

Every method with a frozen backbone—adapter-isolated or linear-probed—has BWT exactly zero on every seed, because nothing shared is updated. Every method that updates the backbone forgets, and the mitigation baselines do not help: EWC ($-0.388 \pm 0.173$), LwF ($-0.564 \pm 0.084$) and replay ($-0.368\pm0.108$) all forget *more* than unregularised fine-tuning ($-0.328 \pm 0.062$). LwF is simultaneously the most accurate method on the first task (0.804) and the worst forgetter, which is the pattern to expect when the distillation term is computed on encoder features that the new task is free to move so long as the task loss outweighs it. At 32 shots, with tasks whose output types differ, none of these constraints is strong enough.

The three-task sequence sharpens this (Table VII). After learning ISLES, full fine-tuning's BraTS Dice falls from $0.785 \pm 0.030$ to $0.231 \pm 0.158$, and the variance is the point: 0.385, 0.295 and 0.013 on the three seeds. A second segmentation task damages the first more than the heterogeneous regression task did, and how much is unpredictable. ISLES

itself then loses 0.19 after brain age. The frozen methods score identically before and after every step, and ISLES Dice is comparable across them (0.437 LoRA, 0.452 linear, 0.417 full fine-tuning), with full fine-tuning again the least accurate—consistent with overfitting 32 small lesions when every parameter is free.

### F. Does the Pretrained Checkpoint Help?

We repeated the LoRA + head ($r$=16) configuration with the backbone randomly initialised instead of loaded from the FOMO checkpoint, changing nothing else. Random initialisation scores Dice $0.787 \pm 0.013$ against $0.777 \pm 0.004$ pretrained—a paired difference of $-0.009$ *against* pretraining (95% CI $[-0.012, -0.007]$, $p = 3{\times}10^{-9}$). On brain age from patches both are at the constant baseline (15.03 vs. 14.72).

We report this plainly: with this checkpoint, on these tasks, at this shot count, pretraining confers no measurable benefit to a frozen backbone, and the small difference runs the wrong way. What the rest of the paper demonstrates is therefore parameter-efficient adaptation of a frozen network—the mechanism does not depend on what the frozen weights contain, and our results should not be read as evidence for transfer from the FOMO corpus. Whether a different checkpoint, a larger backbone, or unfreezing during adaptation would change this is open; a single negative control cannot settle it.

## VI. Discussion

### A. When Is an Adapter the Right Tool?

Our two task types answer this in opposite directions, and the contrast is the useful part.

**When the task head is large, adapters buy economy.** Segmentation inherits a 8.45M-parameter decoder that is already task-shaped; training it is a strong and simple baseline, and LoRA + head matches it to within 0.012 Dice at 90× fewer trainable parameters, closing to equality by 64 shots. Here the argument for adapters is cost—93.8K parameters and under 400 KB per task on disk, against a 34 MB decoder per task—not accuracy.

**When there is no task head to speak of, adapters are necessary.** Brain-age regression attaches a 513-parameter regressor to pooled encoder features. On whole-head input, that head cannot learn the task from frozen features, and neither can bias-only adaptation: both emit a constant. LoRA's 56K encoder parameters can, reaching MAE 10.2 yr, and they do better than releasing all 14M. The representation itself has to move, and low-rank movement is enough—indeed better than unconstrained movement at 32 shots, where full fine-tuning overfits.

**Forgetting-free is a property of freezing, not of adapters.** Adapter isolation and frozen-encoder probing both have BWT zero; the distinction between them is Table I, not the BWT column. What is worth reporting is the other side: none of EWC, LwF or replay achieved it, all forgetting more than plain fine-tuning at this scale, and full fine-tuning's loss on an earlier task of the *same* type ranged from 0.40 to 0.77 Dice across seeds. If prior-task performance matters, freezing is the reliable route and the standard remedies are not.

**The 66-parameter lesson.** Whether the task's output projection is trainable moves Dice by 0.22. This is not a subtle finding, but it is an easy configuration error, and it was the sole cause of a "LoRA underperforms" conclusion in an earlier version of this work.

### B. Limitations

**Pretraining is not shown to help.** The central negative control (Sec. V-F) finds the FOMO checkpoint no better than random initialisation, and slightly worse. Our results support parameter-efficient adaptation of a frozen backbone, not transfer from large-scale pretraining, and the reader should not generalise them to other checkpoints or scales.

**Three seeds, one architecture, one shot count for most results.** Means and SDs are over three training runs; the paired tests pool cases across those runs and test the per-case difference, not run-to-run variability. Only the shot ablation spans 16–64 shots, and only on two methods.

**Task identity is given at inference.** Adapter selection assumes the task is known; task-agnostic routing is not addressed, and it is the obvious next step for a deployed system.

**Two sequences, one order each.** T1→T2 and T1→T3→T2 are the only orders evaluated under the corrected protocol; an earlier reversed-order run was scored against a constant-label defect and is withdrawn.

**Brain age is improved, not solved.** MAE 10.2 yr with $R^2 = 0.40$ beats a constant by a wide margin but is far from the 3–4 yr that large supervised cohorts achieve; 32 labeled subjects and a $96^3$ resampling are the obvious constraints.

**Infarct detection as classification excluded.** A binary infarct-detection formulation contrasting ISLES with IXI controls was prepared but is not reported: the positive class carries DWI, FLAIR and ADC channels and the negative class T1 and T2 with zero-filled channels, so the classes differ by acquisition rather than pathology and any classifier score would be uninterpretable. We use ISLES for lesion segmentation instead, where the cohort is internally consistent.

**No augmentation.** Few-shot results would likely improve with it for all methods; we omit it to keep the method comparison clean.

## VII. Conclusion

We evaluated task-specific LoRA on a frozen 3D U-Net for few-shot continual adaptation across glioma segmentation, stroke-lesion segmentation and brain-age regression, under a subject-grouped held-out protocol with whole-volume inference and three seeds. Whether a frozen backbone with small adapters suffices depends on where the task needs capacity. For segmentation, which inherits a large decoder, LoRA + head trails full-decoder linear probing by 0.012 Dice at 90× fewer trainable parameters and equals it at 64 shots. For brain age from whole-head input, LoRA is the only frozen method that learns the task at all—MAE $10.22 \pm 0.82$ yr against a constant baseline of 14.4—and beats full fine-tuning by 2 years with 56K parameters instead of 14M; the same task from $64^3$ patches is unlearnable for every method, so input scope, not the

TABLE VII

THREE-TASK SEQUENCE T1→T3→T2 (BRATS → ISLES → IXI BRAIN AGE, $64^3$ PATCHES), $N_k = 32$, MEAN ± SD OVER SEEDS 42–44. EACH TASK IS SCORED ON ITS OWN HELD-OUT TEST SET WHEN LEARNED AND AGAIN AFTER EACH LATER TASK. ISLES TEST SET: 155 SUBJECTS.

| Method | At learn | | | BraTS Dice after… | | ISLES Dice |
|---|---|---|---|---|---|---|
| | BraTS Dice | ISLES Dice | Age MAE | ISLES | age | after age |
| LoRA + head, $r$=16 | $0.780 \pm 0.004$ | $0.437 \pm 0.035$ | $14.33 \pm 0.47$ | = | = | = |
| Sequential Linear | $0.789 \pm 0.005$ | $\mathbf{0.452 \pm 0.029}$ | $14.69 \pm 0.12$ | 0.789 | 0.789 | 0.452 |
| Sequential FT | $0.785 \pm 0.030$ | $0.417 \pm 0.022$ | $14.60 \pm 0.06$ | $\mathbf{0.231 \pm 0.158}$ | $0.226 \pm 0.259$ | $\mathbf{0.232 \pm 0.073}$ |

method, was the earlier barrier. Freezing delivers zero backward transfer, which EWC, LwF and replay all fail to do, each forgetting more than unregularised fine-tuning. Finally, the pretrained checkpoint is not better than random initialisation for the frozen backbone, so what we demonstrate is parameter-efficient adaptation rather than transfer. The corrected protocol, splits, per-case predictions and tests are released so that these results can be reproduced and extended.


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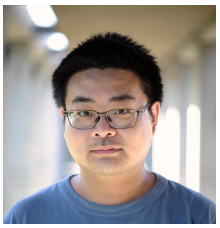
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